

Call for evidence

CCP collateral tokenisation

Responding to this paper

ESMA invites comments on all matters in this paper and in particular on the specific questions summarised in Annex 1. Comments are most helpful if they:

1. respond to the question stated;
2. indicate the specific question to which the comment relates;
3. contain a clear rationale; and
4. describe any alternatives ESMA should consider.

ESMA will consider all comments received by **15 January 2027**.

All contributions should be submitted online under the relevant [consultation](#).

Publication of responses

All contributions received will be published following the close of the consultation, unless you request otherwise. Please clearly and prominently indicate in your submission any part you do not wish to be publicly disclosed. A standard confidentiality statement in an email message will not be treated as a request for non-disclosure. A confidential response may be requested from us in accordance with ESMA's rules on access to documents. We may consult you if we receive such a request. Any decision we make not to disclose the response is reviewable by ESMA's Board of Appeal and the European Ombudsman.

Data protection

Information on data protection can be found at www.esma.europa.eu under the heading '[Legal Notice and Data protection](#)'.

Who should read this paper?

This Call for Evidence is addressed to central counterparties (CCPs), clearing members and their clients, as well as to market infrastructures (CSDs, custodians, triparty agents) and providers of tokenisation or DLT solutions. It is also relevant for legal experts, industry associations and other stakeholders involved in collateral management, post-trade services or digital financial market infrastructure developments.

List of acronyms

API	Application Programming Interface
ART	Asset-Referenced Token
CASP	Crypto-Asset Service Provider
CCP	Central Counterparty
CFTC	Commodity Futures Trading Commission
CPMI	Committee on Payments and Market Infrastructures
CSD	Central Securities Depository
CSDR	Central Securities Depositories Regulation (Regulation (EU) No 909/2014)
DLT	Distributed Ledger Technology
DLT MTF	DLT Multilateral Trading Facility
DLT Regime	Pilot Regulation (EU) 2022/858 on a pilot regime for market infrastructures based on DLT
DLT SS	DLT Settlement System
DLT TSS	DLT Trading and Settlement System
DORA	Digital Operational Resilience Act (Regulation (EU) 2022/2554)
ECB	European Central Bank
EBA	European Banking Authority
EMT	E-Money Token
EMIR	European Market Infrastructure Regulation (Regulation (EU) No 648/2012)

ESMA	European Securities and Markets Authority
FMI	Financial Market Infrastructure
FSB	Financial Stability Board
HQLA	High-Quality Liquid Assets
ICSD	International Central Securities Depository
ICT	Information and Communication Technology
IM	Initial Margin
IOSCO	International Organization of Securities Commissions
MiCA	Markets in Crypto-Assets Regulation (Regulation (EU) 2023/1114)
MiFID II	Directive 2014/65/EU on Markets in Financial Instruments
NFC	Non-Financial Counterparty
OTC	Over-the-Counter
PFMI	Principles for Financial Market Infrastructures
RTS	Regulatory Technical Standards
SFD	Settlement Finality Directive (Directive 98/26/EC)
TR	Trade Repository
VM	Variation Margin

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1 Executive Summary

Reasons for publication

Tokenisation is moving from pilots to early production in a number of areas of wholesale finance. While initiatives remain at different stages of development, a number of public projects (or public-private collaborations) are exploring how DLT-based arrangements could facilitate innovation and further enhance the efficiency of key post-trade processes, including settlement and cross-border multi-currency payments (see in particular ECB Pontes¹ and BIS Project Agora²). Furthermore, a number of industry players are also developing their own tokenisation capabilities and services. In this context, collateral mobilisation has emerged as a practically relevant use case for CCPs. ESMA has therefore decided to further investigate this topic to gain a better understanding of its implications for CCP collateral before deciding whether any action under its remit might be necessary to accompany these developments in a safe and sound regulatory and supervisory framework.

This Call for Evidence seeks stakeholders' views on the emerging use of tokenised collateral in the context of central clearing, in particular whether, and under what conditions, tokenised forms of collateral³ could be used safely and effectively by EU CCPs. The purpose of the exercise is to gather evidence on the legal, market, operational and infrastructure issues raised by tokenised collateral arrangements. The Call for Evidence is not intended to reassess or expand the categories of assets eligible as CCP collateral as such. Rather, it focuses on understanding whether the use of tokenisation may affect the manner in which otherwise eligible assets are mobilised, transferred, managed, protected and realised, and whether existing regulatory and supervisory frameworks remain adequate in that context.

Contents

Following an introduction where ESMA sets the scene on the potential implications of tokenisation in the life cycle of CCP collateral, section 3 presents a high-level picture of the main models which appear to exist for the tokenisation of assets (and cash), section 4 covers legal considerations related to the different use models of tokenisation, section 5 looks at market-risk related aspects such as liquidity and haircuts, section 6 covers questions related to operational resilience and the two remaining sections, 7 and 8, seek views on some transversal issues related to tokenisation as well as other potential use cases.

Next Steps

Following the public consultation on this Call for Evidence, ESMA will consider the feedback it received to this consultation in Q1 2027. Depending on the results of the assessment of the feedback, as well as related work ESMA is carrying out in relation to tokenisation in other areas of financial markets, ESMA will decide on the most appropriate course of action.

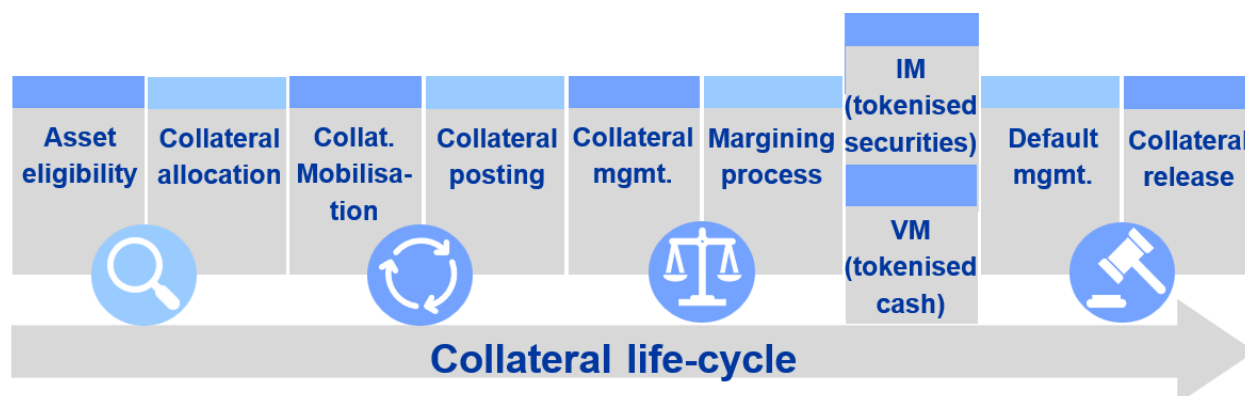
¹ <https://www.ecb.europa.eu/paym/target/pontes/html/index.en.html>

² <https://www.bis.org/about/bisih/topics/fmis/agora.htm>

³ Which can include inter alia DLT financial instruments and tokenised forms of cash, as further developed in this paper.

2 Introduction

1. Collateral plays a central role in central clearing by mitigating counterparty credit risk arising from cleared transactions. In particular, central counterparties (CCPs) collect margin and default fund contributions from their clearing members to cover exposures in the event of a default, thereby contributing to the safety and resilience of financial markets. Under the European Market Infrastructure Regulation (EMIR), collateral posted to CCPs must satisfy strict requirements, including in terms of legal certainty, liquidity, credit quality, valuation, and the ability to be realised in a timely manner.
2. Within this framework, the lifecycle of collateral in central clearing follows a number of well-established stages, from eligibility and allocation, to mobilisation, management and, where necessary, realisation. The diagram below provides a simplified representation⁴ of this lifecycle. It begins with the identification of assets that are eligible for use as collateral and their allocation by clearing members to meet margin and default fund contribution requirements. Once selected, collateral must be mobilised and transferred to the central counterparty (CCP) through the relevant custody, settlement and collateral management arrangements. Following its posting, collateral remains subject to ongoing management processes, including valuation, the application of haircuts, substitution, and monitoring of eligibility requirements.



3. As part of CCPs' risk management process, collateral is collected to cover current and potential future losses linked to exposure arising from a clearing member default. It forms an integral part of the overall margining framework and the management of counterparty risk in central clearing.

⁴ While simplified, this diagram may assist the reader in the understanding of the main aspects of traditional CCP collateral management that could be impacted by tokenisation.

4. Throughout this lifecycle, a range of operational, legal, custody, settlement and risk management processes interact to ensure that collateral can be transferred, managed and, where necessary, realised in a timely and predictable manner, including under stressed conditions and in default scenarios.
5. Given the technological developments experienced in financial markets in the recent years, and in particular those related to distributed ledger technology (DLT), the tokenisation of collateral is gaining momentum among the industry and policy makers⁵. While the purpose of this paper is not to lay down definitions which already exist (in particular in specialised literature, in numerous policy papers or even in legislation⁶), it is however worthwhile reminding what tokenisation means to understand the questions it raises in the context of CCP collateral. As such, tokenisation may be understood as the digital representation of an asset, claim or right, using DLT or similar technologies, such that its transfer and lifecycle events can be recorded and executed in a digital and potentially automated manner. In wholesale markets, tokenisation may take the form either of a tokenised representation of a traditionally held asset (or cash) or of a natively issued tokenised financial instrument or cash.
6. Tokenisation is often associated with efficiency gains, improved asset mobility and greater automation⁷. In the context of central clearing, the operational importance of timely collateral mobilisation, combined with the potential of tokenisation to improve automation and the efficiency of asset transfers, seem to make tokenised collateral one of the most relevant near-term applications of tokenisation⁸. However, tokenisation does not remove the underlying legal, credit, market, liquidity or operational risks associated with the relevant assets or type of transactions. Consequently, tokenisation should not alter the

⁵ In 2025, the ECB's Governing Council has approved a plan that will enable settling distributed ledger technology (DLT) transactions using central bank money. The initiative follows a two-track approach: the first track "Pontes" provides a near-term offering to the market, a Eurosystem DLT-based solution, linking DLT platforms and TARGET Services to settle transactions in central bank money, and the second track "Appia" provides a long term and broader approach for an innovative and integrated ecosystem. In the United States, CFTC staff guidance issued on 8 December 2025 (CFTC Letter No. 25-39 Advisories December 08, 2025) addresses the use of tokenised assets as collateral in derivatives markets and covers, among other aspects, legal enforceability, segregation, custody and control, valuation and operational risks. On the same date, CFTC staff also issued a no-action letter (CFTC Letter No. 26-05 No-Action February 06, 2026) concerning digital assets accepted as margin collateral by futures commission merchants. In the United Kingdom, the FCA and the Bank of England published in May 2026 a [joint shared vision and call for input on tokenisation in wholesale markets](#), expressly covering prudential treatment, settlement arrangements and tokenised collateral, and signalling further policy work in this area. Also, through the Project Agora, the Bank for International Settlements (BIS) has partnered the Institute of International Finance (IIF) to help demonstrate how tokenisation can help improve cross-border payments through multi-currency settlement using tokenised central bank reserves and tokenised commercial bank deposits (<https://www.bis.org/about/bisih/topics/fmis/agora.htm>)

⁶ See [Tokenization and Financial Market Inefficiencies](#), *Fintech Notes No. 2025/001* IMF and recital (3) of Regulation (EU) 2022/858 of the European Parliament and of the Council of 30 May 2022 on a pilot regime for market infrastructures based on distributed ledger technology.

⁷ See *inter alia* preliminary conclusions of the BIS Agora Project, "[Press release: Project Agorá shows how tokenisation can improve wholesale cross-border payments; work will advance to real-value testing](#)"

⁸ See *inter alia* International Organization of Securities Commissions (IOSCO), *Tokenization of Financial Assets*, Final Report FR/17/2025 (2025), pp. 39–40, section F.2, "Collateral Management and Repos."; and, Bank for International Settlements (BIS), "The next-generation monetary and financial system," in *BIS Annual Economic Report 2025* (Basel: BIS, 24 June 2025), Chapter III.

fundamental requirements applicable to CCP collateral, including legal enforceability, settlement certainty, prudent valuation, liquidity and operational resilience.

7. Furthermore, the introduction of tokenised assets for the purpose of collateral raises the question of how the key operational, legal and financial aspects of each stage of the collateral lifecycle described above may be affected when assets or cash are issued, represented, transferred or managed in tokenised form. ESMA is therefore issuing this Call for Evidence to provide a structured basis for assessing how tokenisation may affect existing CCP collateral arrangements taking into consideration legal, risks and operational considerations covered by existing legal frameworks (e.g. EMIR as well as securities laws). The objective is not to assume that tokenisation requires a fundamentally new collateral regime, but rather to assess whether the existing regulatory regime is appropriate for the use of tokenised collateral in the clearing space, or whether tokenised collateral raises material issues in relation to existing EU requirements and supervisory expectations.
8. The analysis should consider both tokenised securities and tokenised forms of cash. Tokenised securities may potentially be used as collateral to satisfy initial margin and default fund contribution requirements, while tokenised cash such as tokenised central bank money or other tokenised settlement assets may also support the settlement of variation margin obligations. Considering both dimensions is important to understand how tokenisation could affect the end-to-end margining process, including collateral mobilisation, settlement, liquidity management, interoperability with existing infrastructures and default management procedures.

Q1. Are you currently using, developing or planning to develop any tokenisation solution relevant to collateralisation central clearing? More generally, are you involved in any tokenisation initiatives, projects or arrangements that could help inform ESMA's understanding of the potential use of tokenised collateral by CCPs? Please describe the initiative and its relevance.

When replying to the questions below, please indicate, if necessary, to which tokenisation solution(s) the answer applies.

3 Operational models

9. Before examining the legal, liquidity and operational resilience issues raised by tokenised collateral, ESMA should form a better understanding of the models which are, or could be, used, tested or developed for tokenised collateral in a CCP context. The purpose of this section is to focus on how those tokenised collateral models would function in practice across the collateral's lifecycle, including its creation or representation, mobilisation, posting, ongoing management, substitution and release, as well as its operational handling in the event of a clearing member default. Legal aspects, including the nature of the

collateral arrangement, client protection, enforceability and settlement finality, are addressed in the following section of this Call for Evidence.

10. For analytical purposes, considering the literature⁹ and the practice (including limited observed examples of tokenised collateral in the central clearing space¹⁰), ESMA understand that tokenised collateral arrangements can broadly be grouped into two implementation models: (i) tokenised representations ("digital twins") of assets that continue to be held and recorded in conventional market infrastructures and (ii) assets issued natively on a DLT platform. However, in practice arrangements may also combine elements of both models. ESMA seeks also views on other potential models which would not be either of these two or the combination of features of both.
11. In a first model, digital twin, the token may evidence, mirror or facilitate the use of rights recorded elsewhere and may or may not itself constitute the legally authoritative record of ownership or entitlement. The token may also be legally constructed as such that the token is constitutive of an individual holding of a pool of securities held with a custodian. In ESMA's view, ensuring consistency between the token layer and the underlying real asset (e.g. "golden record") as well as identifying which record prevails in the event of discrepancy should be one of the central operational features.
12. In a second model, collateral processes would be instead embedded more directly within the ledger-based environment. Assets would be issued, held and transferred only on the relevant ledger¹¹, and operational processes such as posting, monitoring and updating collateral positions would be performed within that environment. In this case, the token would constitute the primary representation of the asset and collateral management would depend on the functioning of the ledger, associated control mechanisms and any linked settlement processes (instead of the double layer system in twin models with a traditional layer plus the DLT layer).
13. Where relevant, the arrangements may interact with tokenised cash (i.e. tokenised central bank money or private money, like tokenised deposits or stablecoins), synchronisation services or other settlement mechanisms, which can determine how collateral movements are finalised or how liquidity is accessed.
14. Across both models, the introduction of tokenisation may affect different stages of the collateral lifecycle to varying degrees. Some processes, such as valuation methodologies or eligibility criteria, may remain largely unchanged as tokenisation should not in itself alter

⁹ See inter alia Kaousar Nassr, I. (2021), "Understanding the tokenisation of assets in financial markets", *OECD Going Digital Toolkit Notes*, No. 19, OECD Publishing, Paris, <https://doi.org/10.1787/c033401a-en>, p. 5.

¹⁰ The industry has developed (or is testing and developing) a number of projects which help issue and/or mobilised collateral through DLT. However, by the time of finalisation of this Call for Evidence, the only project in central clearing that ESMA has seen through its work in EMIR CCP supervisory colleges is [HQLAx](#).

¹¹ ESMA is cognisant that this might include cross-ledger transfers, however, for the sake of the description of the main models not all elements constitutive of each model are detailed.

the prudential quality of the underlying collateral¹². Others, in particular mobilisation, allocation and confirmation of collateral, may be significantly affected, including through increased automation or reduced reliance on traditional settlement steps.

15. An important aspect of the operational functioning of collateral concerns its handling in the event of a clearing member default. From an operational perspective, this involves the sequence of actions through which posted collateral becomes available for use within the CCP's default management process, including its transfer, transformation or preparation for liquidation. While the legal basis for enforcement is addressed separately, the practical effectiveness of this process depends on whether collateral can be accessed and processed, legally and practically, within the required timeframe and without operational impediments. The operational path from posted collateral to liquidity that can be used to manage a default is therefore a key consideration in assessing tokenised collateral arrangements
16. In digital twin models, according to ESMA's understanding based on early experience, this operational path is expected to remain closely linked to the underlying traditional infrastructure. Although the token layer may facilitate the identification, allocation and pre-positioning of collateral, the ability to use or dispose of the asset upon default is likely to depend on processes within custodial or CSD systems. This may require coordination between the token layer and the underlying infrastructure, including reconciliation, confirmation and messaging steps, before the collateral can be mobilised for sale, repo or other forms of realisation. The extent to which these additional steps introduce delays or dependencies is likely to be a key determinant of operational effectiveness in stress scenarios
17. In native tokenised asset models, the operational pathway to value would seem to be more directly linked to the ledger environment. Collateral should be transferred, sold or otherwise used through on-ledger processes, or through interaction with settlement assets and payment mechanisms connected to the platform. In such arrangements, the timing and predictability of the process would probably depend on the availability of liquidity, the functioning of settlement mechanisms and any reliance on external infrastructures or intermediaries.
18. In ESMA's views, the ability to convert the tokenised asset into usable funds, and the dependencies associated with that process¹³, are therefore central to the assessment of how such models would perform in both normal and stressed conditions.

Q2. Does the taxonomy used by ESMA in this Call for Evidence (“digital twin”, “native”, “hybrid”) correspond to the practice or are there other “models” to consider? How do

¹² Cf. discussion on liquidity in section 5 of this paper.

¹³ In that context, Eurosystem's Pontes, by connecting DLT-based solutions (incl. DLT market infrastructures) with TARGET Services, could play a pivotal role for tokenized collateral while preserving the role of central bank money in the financial system as the safest means of settlement.

you see the evolution of the models and which role you envisage for tokenised collateral/assets in clearing more generally?

Q3. Is there any specific preferred model for tokenised collateral in the clearing space? If yes, how would this impact the existing regulatory framework?

Q4. Depending on the preferred model (or models), how would tokenised collateral be handled operationally in a default scenario? Please describe the main steps from posted collateral to liquidity realisation and including any key bottlenecks or dependencies.

4 Legal considerations: Legal certainty, segregation, insolvency implications and settlement finality

19. For collateral to be effective in central clearing, it must be supported by a clear and robust legal framework that ensures legal certainty in normal conditions and in the event of a clearing member default. Also, a token may qualify as a financial instrument, but this does not necessarily mean that a DLT record alone is sufficient to create, transfer or protect ownership rights under national law. As property, securities and insolvency laws remain largely national, the legal effects of tokenisation should be assessed separately from the legal qualification of the token itself. This includes clarity on the legal nature of the collateral arrangement, the treatment of collateral in insolvency, the protection of client assets, the enforceability of rights, and the extent to which transfers of collateral are legally final and irrevocable.
20. ESMA is carrying out analytical work with national competent authorities to understand whether and how national legal frameworks have evolved to take account of the emergence of DLT in financial markets and whether barriers to the development of tokenisation exist. The objective of this section of ESMA's Call for Evidence is to gather further evidence on the legal implications of tokenised collateral arrangements to complement ESMA's internal work to identify (and address as relevant) legal questions and concerns related to the use of tokenised collateral.
21. Taken together, the three dimensions analysed in this section (the legal nature of tokenised collateral arrangements, the effectiveness of segregation and client asset protection, and the conditions for settlement finality and irrevocability) are closely interdependent and jointly determine the legal robustness of tokenised collateral frameworks in a CCP context. Each element addresses a different aspect of the same requirement: that collateral must confer clear, enforceable and insolvency-protected rights, which can be reliably identified, controlled and realised without legal or operational uncertainty.
22. In traditional post-trade infrastructures (although some legal risks and frictions may still exist), these outcomes are supported by established legal concepts, including title transfer and security interests, and by regulated intermediaries operating within statutory

frameworks that provide legal certainty with respect to settlement finality and the segregation of client assets. In tokenised arrangements, by contrast, some functions may be performed or supported through technological infrastructure, governance mechanisms and contractual arrangements, requiring careful consideration of how legal rights and obligations are created, evidenced and enforced¹⁴.

23. The central question is therefore not whether tokenisation changes the economic function of collateral, but whether it could preserve the legal certainty, proprietary clarity and insolvency resilience that underpin existing CCP collateral frameworks. This includes ensuring that rights over collateral are consistently characterised under applicable law, that client assets are effectively segregated and protected throughout the custody and settlement chain, and that transfers of collateral reach finality in a manner that is both operationally reliable and legally irrevocable.

4.1 Legal nature of the collateral arrangement

24. A first key issue concerns the legal nature of tokenised collateral arrangements. In traditional CCP frameworks, collateral is typically provided either through title transfer arrangements or through security interests, such as pledges, charges, or equivalent mechanisms. Under a title transfer structure, the CCP becomes the legal owner of the collateral upon posting, subject to an obligation to return equivalent assets if no default occurs. Under a security interest structure, ownership remains with the collateral provider, while the CCP acquires enforceable rights over the collateral that may be exercised upon the occurrence of a default.
25. The introduction of tokenised collateral raises the question of whether these legal structures can be replicated effectively in a distributed ledger environment and whether the tokenisation of an asset (be it a digital twin or a DLT-native asset) affects the legal characterisation of the arrangement. Depending on the technical characteristics of the DLT as well as on any potential national legal framework applicable to securities represented in DLT, the transfer of a token may or may not, by itself, determine the transfer of proprietary rights in the underlying asset or the creation, perfection or enforceability of a security interest. Likewise, technical control over a token or the associated private keys might not mean legal ownership or legally recognised control, as these legal effects ultimately depend on the applicable legal and contractual framework¹⁵.

¹⁴ See inter alia CPMI, *Tokenisation in the Context of Money and Other Assets: Concepts and Implications for Central Banks*, October 2024, discussing how tokenisation reallocates functions traditionally performed by intermediaries across technological, governance and legal arrangements.

¹⁵ UNIDROIT, *Principles on Digital Assets and Private Law*, 2023, in particular Principles 5, 6, 8 and 14–17 and the accompanying Commentary. See especially the Commentary explaining that the legal effect of a link between a digital asset and another asset, as well as the validity and third-party effectiveness of transfers and security rights, are matters governed by the applicable law ("other law"). See also CPMI, *Tokenisation in the Context of Money and Other Assets: Concepts and Implications for Central Banks*, October 2024, Sections 2–4, on the relationship between token transfers and the legal rights attached to tokenised assets.

26. Accordingly, a key question is whether a transfer of tokenised collateral constitutes: (i) a transfer of ownership, (ii) the creation or perfection of a security interest over that asset, or (iii) merely a transfer of a digital representation that does not itself confer proprietary rights. Legal certainty requires that the rights acquired by the CCP through the transfer or encumbrance of the token correspond clearly and unequivocally to enforceable rights in respect of the asset.
27. This issue becomes particularly relevant where tokenised collateral would be held through multiple layers of infrastructure. Depending on the design of the arrangement, rights may be recorded simultaneously in a distributed ledger, in the books of a custodian, in a traditional securities settlement system, or in other dedicated records¹⁶. In some instances, such structures may raise questions regarding the location of the asset, the identity of the person recognised as holding proprietary rights, and the hierarchy of claims among participants and intermediaries. The legal treatment of tokenised assets could therefore differ depending on whether the tokenised asset would exist natively on a distributed ledger or would merely constitute a tokenised representation of an asset held in a conventional infrastructure.
28. In addition, while recognising that this might also happen in a more traditional setting, tokenised collateral arrangements may give rise to complex conflict-of-laws issues, particularly in cross-border settings. This issue may be particularly acute where the issuer, custodian, tokenisation platform, validator nodes, clearing member, CCP and clients are located in different jurisdictions, or where the applicable law of the token, the law governing the proprietary effects of the ledger entries or applicable insolvency laws differ from the law governing the underlying asset or collateral agreement. Traditional collateral arrangements usually rely on well-established frameworks to determine the applicable law governing proprietary aspects of collateral arrangements, such as the law of the account, the location of the intermediary, or the location of the asset. In a distributed ledger environment, where records may be maintained simultaneously across multiple jurisdictions, the determination of the applicable law may be less straightforward. Legal certainty requires that parties can identify with confidence the law governing ownership, perfection, priority, and enforcement of rights in tokenised collateral.
29. For CCPs, the critical question is how to ensure that the legal rights obtained over tokenised collateral are equivalent in certainty, enforceability, and effectiveness to those obtained over traditional collateral. Any ambiguity regarding ownership, the creation or perfection of security interests, or the recognition of proprietary rights could undermine the ability of a CCP to realise collateral promptly and effectively in a default scenario. Consequently, it is essential to understand how tokenised collateral arrangements are structured in practice, whether existing legal frameworks adequately accommodate such arrangements, and whether legislative or contractual adaptations may be necessary to

¹⁶ E.g. In Spain, a law enacted in 2023 introduced provisions requiring a designated responsible entity (ERIR) to administer and validate the decentralized registries used for recording and tokenizing financial instruments.

ensure legal certainty across different jurisdictions and infrastructures. A related question is whether existing legal frameworks sufficiently recognise control over tokenised collateral for the purposes of title transfer, pledge, enforcement and close-out. In particular, whether market participants rely on statutory recognition, contractual arrangements, account-based records, control of private keys or a combination of these elements to establish enforceable rights.

30. The Financial Collateral Directive (FCD) requires that financial collateral must “*be in the possession or under the control of the collateral taker or of a person acting on the collateral taker’s behalf*” (Article 2(2) FCD). Accordingly, where the FCD applies, the relevant requirements concerning would need to be capable of being satisfied in a DLT environment. Although the FCD is principally concerned with bilateral collateral arrangements, this does not mean that it is confined to collateral arrangements outside financial market infrastructures: CCPs, settlement agents and clearing houses are expressly included among the entities that may be parties to an FCD arrangement. The FCD may therefore be relevant to collateral provided by clearing members or other eligible counterparties to CCPs, including where such arrangements are situated within a wider multilateral system, subject to the applicable requirements of the FCD and, where relevant, the SFD.

Q5. How are (or should be) tokenised collateral arrangements structured from a legal perspective (for example in terms of title transfer or security interest), and do current frameworks (mostly embedded in national laws) provide sufficient clarity as to the nature of the rights involved?

Q6. To what extent tokenised collateral arrangements ensure that the CCP can exercise its rights over collateral effectively and without delay in the event of default?

Q7. Are there specific legal uncertainties or barriers, including in cross-border contexts, that may affect the enforceability of tokenised collateral?

Q8. How is the applicable law governing property rights determined in relation to tokenised collateral arrangements? Is it determined the same way as for traditional collateral arrangements?

Q9. To what extent do you see legal uncertainties regarding the requirement of possession or control under the FCD? Does control of a wallet or private key have any legal significance for the creation, perfection or enforcement of collateral rights, and if so under which legal framework?

4.2 Segregation and client protections

31. A key question in assessing the viability of tokenised collateral arrangements is whether they can deliver outcomes equivalent to existing segregation and client asset protection

requirements under EMIR, while accommodating the operational and technological specificities of distributed ledger and tokenised infrastructures.

32. In traditional post-trade arrangements, segregation is typically achieved through account-based custody chains maintained by custodians, CSDs or triparty agents. These account structures establish a clear legal and operational linkage between the recorded assets, the relevant intermediary, and the beneficial owner or client. They support key regulatory and legal objectives, including the identification and separation of client assets, the portability of positions and collateral in the event of a clearing member default, and the protection of assets in insolvency scenarios, including under the FCD and the Settlement Finality Directive.
33. In tokenised or DLT-based environments, these functions traditionally performed through account-based custody arrangements may no longer rely exclusively on conventional custody chains. Instead, equivalent outcomes may be achieved through a combination of ledger architecture, permissioning mechanisms, governance arrangements and cryptographic authentication mechanisms that regulate access to, and the transfer of, tokenised assets, while remaining dependent on the applicable legal and contractual framework¹⁷. As a result, segregation may need to be understood less as a purely account-based legal construct and more as a set of functional requirements that must be achieved regardless of technological implementation. These functional requirements typically include the ability to:
- clearly attribute assets to specific participants or clients in a manner that is legally and operationally recognisable;
 - prevent unauthorised use, rehypothecation, or double use of the same assets; and
 - ensure that such attribution and control remain effective, reliable, and enforceable, including in stress scenarios such as default or insolvency.
34. In a tokenised collateral environment, the question arises of how segregation can be legally achieved, evidenced, and enforced. It is relevant to assess whether segregation is achieved at the level of the token (e.g. through wallet-level separation, smart contract-based controls, or permissioned ledger structures), at the level of the underlying asset, or through a combination of both layers. Technical separation through wallets, permissions or smart contracts may support segregation, however they might not necessarily imply legally enforceable attribution of assets, exclusion from the insolvency estate of custodians or platforms operators, prevention of double use. ESMA's interest in this respect is whether such technical mechanisms can reliably replicate the legal effects of segregation under existing EU regulatory frameworks.

¹⁷ See UNIDROIT *Principles on Digital Assets and Private Law*, 2023, in particular Section III on control and transfer, <https://www.unidroit.org/wp-content/uploads/2024/01/Principles-on-Digital-Assets-and-Private-Law-linked-1.pdf>

35. At the same time, tokenised collateral arrangements may increase structural complexity, particularly where tokenised representations coexist with underlying assets held within traditional infrastructures. In those cases, effective segregation requires not only internal consistency within the tokenised system, but also robust legal and operational reconciliation between the token layer and the underlying “golden record”. Any divergence between these layers may create risks of double pledging, inconsistent records of ownership or control, or legal uncertainty as to which record prevails in insolvency or enforcement scenarios.
36. In such structures, the allocation of responsibilities between custodians, tokenisation platform operators, DLT nodes, and other intermediaries becomes particularly relevant. This includes clarity on who bears legal responsibility for maintaining accurate records, ensuring segregation of client assets, and preventing unauthorised transfers or reuse. Governance arrangements governing access rights, validation of transactions, and reconciliation processes may also be central to ensuring that segregation is not only technically implemented but also legally effective.
37. From a legal perspective, ESMA understands that the effectiveness of asset segregation in a tokenised environment would depend on the proprietary characterisation of the rights associated with the token, as well as on the legal structure through which collateral would be provided, including whether the arrangement takes the form of a title transfer or a security interest¹⁸. Where token holders do not benefit from clearly defined and enforceable proprietary rights, or where segregation mechanisms fail to ensure effective separation of client assets, there is a risk that participants may be treated as unsecured creditors in insolvency proceedings. This risk is particularly acute in multi-layered arrangements involving intermediaries across different jurisdictions or legal regimes.
38. Segregation arrangements should also be assessed from the perspective of portability. In a clearing member default, client positions and related collateral may need to be transferred to a back-up clearing member within a short timeframe. Tokenised collateral arrangements should therefore be assessed not only by reference to asset protection in insolvency, but also by reference to whether they support effective porting, including where wallets, permissions, platform access or identity controls are involved. Interoperability (or interconnectivity) may play an important role here.
39. The assessment of segregation cannot be limited to operational design alone, but must also consider whether the underlying legal framework ensures that segregation is recognised and upheld in insolvency, enforcement, and conflict-of-laws scenarios.

¹⁸ For further discussion on legal control and transfer of tokenised financial instruments see UNIDROIT, *Principles on Digital Assets and Private Law*, 2023, especially Principles 5–8 and 14–17 and the accompanying Commentary, discussing the proprietary characterisation of digital assets and the creation and effectiveness of security rights.

40. Against this background, ESMA seeks evidence on how tokenised collateral arrangements could implement segregation in practice, including how client assets are distinguished, controlled, and protected across different operational models, and whether existing EU requirements on segregation and client asset protection can be effectively mapped onto tokenised infrastructures or would require further legal or regulatory clarification.

Q10. How are, or should be, segregation and client protection requirements implemented in tokenised collateral arrangements, and does tokenisation affect the level of protection provided compared with conventional models? Please distinguish technical separation from legal separation.

Q11. Do tokenised collateral arrangements support the timely porting of client positions and collateral in a clearing member default? What operational or legal dependencies could affect portability?

4.3 Settlement finality and irrevocability of collateral transfers

41. CCP collateral arrangements must ensure that transfers of collateral, or the acquisition of control over collateral, are final and irrevocable in a legal sense. This requirement is central to the effectiveness of collateral in risk management. It ensures that once assets have been provided to a CCP, they cannot be unwound, reversed, or otherwise challenged at a later stage, including in the context of insolvency proceedings affecting a clearing member, an intermediary, or another participant in the collateral chain.

42. From a legal perspective, settlement finality constitutes the legal point at which a transfer becomes definitive, no longer conditional, and no longer exposed to retrospective disruption. In modern financial markets, where transactions are processed through interconnected infrastructures involving CCPs, CSDs, custodians and payment systems, this legal certainty is essential to the stability of the system as a whole. Without a clearly defined and enforceable point of finality, exposures that are assumed to be extinguished could re-emerge after the fact, undermining margining, liquidity planning and default management.

43. Within the EU framework, these objectives are reflected in the protections afforded by the SFD, which limits the effects of insolvency law in relation to transfer orders, netting and collateral in designated systems. The framework allows the relevant system rules to determine key moments such as entry into the system and irrevocability, while ensuring that, once those moments are reached, subsequent insolvency proceedings cannot unwind the relevant transactions. This interaction between system rules and legal protection is central: finality emerges not from a technological feature which seems to result in the transfer of the entitlement in the collateral from one party to another, but from the legal recognition of specific operational events as definitive which may or may not coincide with technical settlement finality.

44. In tokenised and DLT-based environments, the concept of finality raises a number of additional interpretative and practical questions. Limited experience observed by ESMA through its internal work with national competent authorities (largely based in projects authorised through the DLT Pilot Regime) shows that currently these projects have implemented finality-like protections through private law contractual clauses. While these projects do not concern CCP collateral management, they represent interesting experience. However, a broader assessment is needed to better understand the implications of finality on DLT at greater scale where the designation of a system under SFD could become crucial for finality protection.
45. A key issue is the relationship between technical finality and legal finality. In traditional infrastructures, these two dimensions tend to be closely aligned: system rules define when a transfer becomes irrevocable, and the law recognises that moment as legally final. Understanding how this would work in a tokenised environment is key.
46. It is ESMA's understanding that certain technical and governance features of tokenised systems may affect how technical finality of the transfer is achieved in practice. These features may include mechanisms allowing transaction reversal or amendment under exceptional circumstances, administrative override functions, smart contract logic enabling conditional settlement or unwind, or continued reliance on off-ledger processes such as reconciliation with custodial or CSD records.
47. From a CCP perspective, it is therefore critical that the moment at which collateral is considered to be posted and available for risk management purposes coincides with a point at which the transfer or control of that collateral is legally robust, irrevocable, and protected against subsequent challenge. Any misalignment between operational confirmation and legally recognised finality may create timing risk, unexpected exposure or uncertainty in default management processes. This is particularly acute in stressed conditions, where large volumes of transactions are processed under time constraints and where legal certainty is most needed.
48. Against this background, the question is whether tokenised collateral arrangements can provide a level of legal certainty and alignment between operational and legal finality that is equivalent to that achieved in existing settlement systems. This includes considering whether current legal concepts and system-based approaches to finality can be effectively applied in tokenised environments, or whether additional clarification is required to ensure that the legally relevant moment of finality remains clear, consistent and enforceable across different technological and institutional configurations.

Q12. How is the point of finality defined in tokenised collateral arrangements, and how does it relate to both the technical processing of transactions and the legal framework applicable to the system?

Q13. Are there features of tokenised infrastructures or governance arrangements that could adversely affect or on the contrary enhance/support the irreversibility of collateral transfers (and therefore the finality)?

Q14. In arrangements involving multiple layers (such as tokenised systems interacting with traditional infrastructures), how could finality be ensured, including consistency between operational records and legally recognised outcomes?

5 Liquidity and haircuts

49. A key element of CCP risk management is the ability to rely on collateral not only in normal market conditions, but also under stress and in the event of a clearing member default. Under EMIR, CCPs are required to accept highly liquid collateral with minimal credit and market risk and to apply haircuts that reflect the potential for the value of the collateral to decline over the interval between its last revaluation and the point at which it can reasonably be assumed to be liquidated. This assessment differs in nature depending on whether collateral is provided in the form of securities or cash.
50. For securities, liquidity risk primarily relates to the ability to sell, repo or otherwise realise the asset without significant price impact within the relevant liquidation period, including in stressed market conditions. Haircuts applied to such assets are therefore typically designed to cover potential adverse price movements, as well as liquidity and concentration risks that may arise during the liquidation process.
51. For cash, liquidity is generally less dependent on market depth or sale conditions, but may still give rise to risks linked to currency mismatch, convertibility, credit exposure to the issuer or deposit-taker, and the operational process required to access and transfer funds. For this reason, CCPs may apply measures analogous to haircuts or other valuation adjustments, in particular where cash is denominated in a currency different from the one in which exposures are measured, or where the availability of funds depends on specific intermediaries, infrastructures or redemption mechanisms. In both cases, the relevant consideration is the extent to which collateral can be made available in a timely, predictable and reliable manner to meet obligations arising in the margining and default management process.
52. In a tokenised context, an initial question is whether the tokenised form of an asset modifies this assessment, both for securities and for cash. In principle, tokenisation does not alter the economic nature of the underlying asset and should not, in itself, justify a different prudential treatment. Accordingly, a tokenised government bond or share should, as a starting point, reflect the same credit and market risk characteristics as its traditional equivalent, and a tokenised cash instrument should, in principle, reflect the value of the underlying monetary claim. However, tokenisation changes the way in which assets are

transferred, accessed and ultimately converted into liquidity that can be used to manage a default, and this is important in a CCP context.

53. For tokenised securities, the key question is whether the tokenised form alters the time, process, certainty, market liquidity or security with which the asset can be sold or repoed, including in time of stress.
54. In this respect, even where the tokenised asset is highly liquid in traditional form, its tokenised form could still potentially introduce liquidity risks or other risks (e.g. conversion, operational, potential loss from cyber risk). These could arise, for example, if access to cash depends on redemption processes, if transfers require interaction with specific platforms or settlement assets, if the tokenised security is subject to constraints on convertibility or transferability, the degree of connectivity of the platform with the broader financial ecosystem, or the reliance of public infrastructures. Such features are relevant when assessing the effective liquidity and risk of the collateral and, where appropriate, the eligibility, level of haircuts or other adjustments needed to reflect the time and certainty with which value can be realised.
55. For cash collateral, liquidity considerations are generally different from those applicable to securities. Cash collateral can normally be used directly to meet obligations and does not need to be liquidated through a sale or repo transaction. However, the effectiveness of cash collateral may still depend on the form in which it is held or transferred, including whether it is available in central bank money, commercial bank money, or through another arrangement that gives rise to a claim on an issuer or deposit-taker. In tokenised environments, a number of settlement assets could be used, including tokenised central bank money, stablecoins, e-money tokens, or tokenised deposits. Some of these assets however, may not be considered equivalent to cash by regulatory and/or supervisory frameworks, in particular stablecoins in view of potential deviation from par. Furthermore, their effective liquidity may depend on whether the chosen asset can legally discharge cash obligations, whether the parties agree to treat it as being in lieu of a payment, the conditions under which they can be converted into settlement money, including the reliability of redemption rights, the credit quality of the issuer, the availability of conversion mechanisms and the operational robustness of the relevant infrastructures. These features may be particularly relevant where exposures need to be covered in a specific currency or within a short timeframe, or where access to liquidity depends on specific intermediaries or platforms.
56. Finally, the assessment of tokenised collateral (whether securities or cash) should also consider whether tokenisation introduces new forms of wrong-way risk or concentration risks.
57. These considerations also have implications for the selection of tokenised assets as eligible collateral. For securities, the benchmark remains the availability of deep and resilient secondary and repo markets, which allow assets such as government bonds to be

converted into cash quickly and with limited price impact, including under stress. For tokenised cash instruments, the benchmark is the ability to reliably access cash in the relevant currency, with minimal delay and uncertainty.

Q15. For all assets already eligible as CCP collateral, what aspects of their tokenised form would affect the time, process or certainty with which they can be converted into cash, including under stressed conditions? Please explain the factors driving such effects, whether they are expected to be positive or negative, whether they depend on specific operational or technical arrangements, and whether any safeguards or adjustments (including to haircut methodologies) may be warranted.

Q16. Do tokenised collateral arrangements give rise to specific forms of wrong-way risk or concentration risk, in particular where their value or liquidity depends on specific issuers, platforms or counterparties? If so, how should such risks be identified and reflected in collateral eligibility, liquidity assessment or haircut calibration?

Q17. What factors are most relevant for assessing the liquidity of tokenised collateral, including for both securities and tokenised cash?

58. Beyond the characteristics of individual assets, tokenisation may also affect liquidity risk at a more systemic level. Faster settlement cycles, increased automation or programmability may alter the timing of collateral flows and margin calls. While such features may improve efficiency and reduce certain types of operational friction, they may also increase intraday liquidity demands by shortening the time available for participants to source eligible collateral or settlement assets.

59. In addition, tokenised arrangements may facilitate the use of automated or conditional processes, including the automatic mobilisation, substitution or transfer of collateral when predefined conditions are met. While such mechanisms may increase efficiency, they could also result in more synchronised behaviour during periods of market stress. For example, if multiple participants are subject to similar triggers, collateral calls or transfers may occur simultaneously across a large number of market participants, potentially increasing demand for the same assets or sources of liquidity at the same time. This could amplify procyclical dynamics, particularly in stressed market conditions, where participants may need to obtain liquidity or eligible collateral within increasingly compressed timeframes.

60. These effects are relevant for both securities and cash collateral and may need to be considered when assessing liquidity risk and calibrating haircuts.

Q18. Do tokenised arrangements (e.g. faster settlement or automation, automation, programmability or continuous processing) have broader implications for liquidity risk, including intraday needs or procyclicality?

61. EMIR imposes requirements not only on the eligibility of collateral accepted by CCPs, but also on how such collateral is held, managed and invested. In particular, CCPs must

ensure that collateral remains sufficiently liquid, secure and readily available, including under stressed market conditions. In this context, tokenisation may raise questions as to whether the tokenised form of otherwise eligible assets affects the ability of CCPs to manage and invest collateral in accordance with Article 47 of EMIR.

Q19. For assets already eligible as CCP collateral, would tokenisation affect the ability of CCPs to manage and invest collateral in accordance with EMIR requirements? Please explain.

6 Operational resilience

62. Operational resilience is a core requirement for CCPs. CCPs must maintain and operate an organisational structure that ensures continuity and orderly functioning in the performance of its services and activities. This requires robust governance, effective identification and management of operational risk, clear allocation of responsibilities, sound change management, business continuity arrangements, recovery capabilities and regular testing of critical systems and processes. In a CCP context, operational resilience is not limited to the availability of technical systems in a narrow sense; it also concerns the ability of the CCP to continue collecting margin, managing collateral, processing payments and carrying out default management functions within the timeframes required for risk management.
63. In addition, Regulation (EU) 2022/2554 on digital operational resilience for the financial sector ("DORA") further strengthens the operational resilience framework applicable to CCPs by introducing harmonised requirements in relation to ICT risk management, incident detection and reporting, operational continuity and the management of ICT third-party risks. In particular, CCPs are required to ensure that their critical systems and supporting infrastructures are resilient to disruptions, including cyber-related risks, and that dependencies on external service providers do not undermine the continuity of core clearing functions.
64. The use of tokenised collateral does not change these operational resilience expectations. What tokenisation may change is the way in which operational and cyber risks arise, and the channels through which those risks may affect the collateral process including whether a DLT is permissioned or permissionless. For example, in a digital twin model, the underlying asset may remain safely held within a traditional custody or CSD chain, but an outage, coding error or reconciliation failure in the tokenisation layer could still prevent the collateral from being mobilised, confirmed or made available to the CCP when needed. In a native token model, a disruption affecting node operations, key management, permissions or smart contract execution could prevent collateral from being transferred or used even where the asset itself remains valid. Similarly, cyber incidents affecting administrative privileges, interfaces or wallet infrastructure could impair collateral

movements or create uncertainty as to the state of the system at a critical moment, including during a default.

65. Operational risks are already present in traditional infrastructures. However, DLT-based and other tokenised arrangements may modify their profile by introducing additional dependencies across technology components, network participants and governance arrangements (in particular in permissionless DLTs if network participants and governance arrangements are not controlled by an identifiable entity, but decentralised and potentially anonymous). Tokenised collateral processes may require the coordinated functioning of smart contracts, permissioning mechanisms, settlement infrastructures and conventional custody arrangements, increasing the importance of interoperability and operational resilience across both tokenised and traditional layers¹⁹.
66. A particular issue is the identification of components of tokenised collateral arrangements that should be regarded as critical dependencies for supervisory purposes. In practice, this would generally include any component whose failure could prevent the CCP from receiving, recording, verifying, managing or using collateral within the required time horizon. Depending on the operational model, this may include the ledger or tokenisation platform itself, smart contracts or other logic governing collateral status, key management and access control arrangements, interfaces with custodians, CSDs, triparty agents or collateral management systems, reconciliation and messaging layers, settlement links, synchronisation mechanisms and, where relevant, the arrangements used for tokenised cash or settlement assets.
67. Testing and assurance are of central importance. This may require a combination of technical testing and process testing. Testing would encompass technical aspects (including code review, smart contract assurance, security testing, failover testing and recovery testing) and operational (test end-to-end workflows involving the CCP and other parties involved such as custodians and technology vendors). Testing should also include default management simulations, cyber-incident scenarios, loss or compromise of private keys, platform outage, failed reconciliation between token and underlying records, smart contract malfunction, governance intervention, and stressed conversion of tokenised collateral into usable liquidity.
68. The resilience assessment should also take account of the fact that tokenised collateral arrangements may rely on a relatively small number of specialist providers or shared infrastructures. Third-party concentration and outsourcing risks may therefore become particularly relevant. This may arise, for example, where multiple CCPs or market participants depend on the same tokenisation platform, cloud environment, smart contract

¹⁹ See *inter alia* CPMI, *Distributed Ledger Technology in Payment, Clearing and Settlement: An Analytical Framework*, February 2017, discussing operational, governance and interoperability risks arising from DLT-based arrangements; IOSCO, *Tokenisation of financial assets*, November 2025, discussing operational, cyber, governance and third-party dependency risks; and CPMI, *Tokenisation in the Context of Money and Other Assets: Concepts and Implications for Central Banks*, October 2024, discussing the interaction between tokenised arrangements and existing financial market infrastructures

developer, connectivity provider, wallet infrastructure, or other specialist service provider. In such cases, operational resilience depends not only on the robustness of the arrangement at entity level, but also on the substitutability of key services, the consequences of a service outage for multiple users, and the ability to maintain operations if a critical provider fails or becomes unavailable.

69. Overall, in ESMA's views, the key question is whether tokenised collateral arrangements can deliver an operational resilience outcome that is at least equivalent to that expected in existing collateral arrangements relied upon by CCPs. This includes not only secure and reliable technology, but also robust governance, clear accountability, strong controls around changes and upgrades, realistic testing and assurance, well-defined fallback arrangements, and a clear understanding of operational dependencies within the broader collateral and settlement chain. Clarifying these issues is essential in order to assess whether tokenised collateral can be used safely and at scale in a CCP environment.

Q20. Does this description accurately capture the main operational and cyber risks associated with tokenised collateral in a CCP context? If not, which additional risks should be considered? Please elaborate detailing the taxonomy of tokenised-specific risks that should be incorporated into CCP risk management approaches.

Q21. Which components of tokenised collateral arrangements should be regarded as critical dependencies for supervisory purposes, and why?

Q22. What forms of testing, assurance (including but not limited to potential fallback arrangements) and scenario analysis (in particular potential new extreme but plausible scenarios) would be necessary before tokenised collateral arrangements could be relied upon more broadly by CCPs?

7 Transversal issues

70. The preceding sections have examined the main legal, operational and market aspects of tokenised collateral in a CCP context. Those sections focus on issues that go directly to the viability of tokenised collateral as a financially sound form of collateral. There are, however, a number of broader or transversal issues that may also prove important for the safe development of tokenised collateral arrangements. These issues may shape whether tokenisation remains limited to isolated use cases or can develop in a way that is scalable, resilient and consistent with the functioning of the wider clearing ecosystem.

Interoperability and fragmentation

71. A first transversal issue concerns interoperability²⁰. Some of potential benefits of tokenised collateral include faster collateral mobility, reduced operational friction and more efficient

²⁰ Not to be confused with interoperability for CCPs under Title V of EMIR.

use of eligible assets. However, for those benefits to materialise, tokenised collateral arrangements cannot develop as separate or closed systems that do not interact effectively with all the relevant stakeholders involved in the clearing chain. The lack of interoperability across tokenisation market infrastructures has been identified as one of the key structural constraints on scaling tokenised finance²¹.

72. In a CCP context, this issue is particularly important because collateral is valuable only if it can be mobilised and used where it is needed, without unnecessary delay, friction or dependency on a single infrastructure. If an asset is already eligible collateral in traditional form, there is a strong argument that its tokenised form should not become operationally captive to one venue, one collateral platform or one infrastructure arrangement. A lack of interoperability at any one of the relevant layers may be sufficient to prevent tokenised collateral from scaling safely or efficiently. This is also relevant in the assessment of the liquidity and market profile of the tokenised asset.

Q23. Does this description accurately capture the main interoperability challenges associated with tokenised collateral in a CCP context? In particular, would a lack of interoperability risk fragmenting tokenised collateral across platforms, infrastructures or CCPs?

Q24. Are common market practices, technical standards or regulatory approaches likely to be needed to support interoperability and avoid the emergence of closed or fragmented tokenisation models?

Market structure and concentration

73. A second transversal issue concerns the potential impact of tokenisation on market structure. Even where tokenisation improves individual processes, it may also create new dependencies or shift market power towards other stakeholders such as platforms, technology providers, private governance arrangements that may become critical to the functioning of tokenised infrastructures (e.g. entities controlling the rules of a permissioned platform, foundations, consortiums or committees governing a network) or service providers (e.g. providers of interoperability or synchronisation services, custodial or wallet-service providers). International work on tokenisation has pointed to increased interconnectedness and dependencies as one of the structural challenges that may emerge as tokenised markets scale.

74. From a CCP perspective, concentration may matter both at the level of operational resilience and at the level of market functioning. If tokenised collateral can only be mobilised efficiently through a limited number of infrastructures or governance models, this could increase switching costs, reduce substitutability, and create bottlenecks or critical

²¹ See inter alia IOSCO, *Tokenisation of financial assets*, November 2025, in particular the structural challenges identified in the conclusion of this report.

points of dependency in stress scenarios. The question is therefore not only whether tokenisation is technically feasible, but also whether its development could reshape the ecosystem for collateral management in ways that create new forms of concentration or dependence.

75. A further transversal issue concerns the allocation of regulatory responsibilities across the tokenised collateral chain. Tokenised collateral arrangements may involve CCPs, CSDs, custodians, settlement systems, tokenisation platforms, wallet providers, technology vendors, issuers of tokenised cash instruments and other intermediaries. Depending on the model, different EU frameworks may become relevant, including EMIR, CSDR, the Settlement Finality Directive, the Financial Collateral Directive, MiCA, DORA and, where applicable, the DLT Pilot Regulation.

Q25. Could tokenisation lead to new market structure dependencies or concentration risks? If so, in what form?

Q26. How do you expect market structure to be impacted by broader usage of tokenised collateral? What sort of risks could be envisaged? How would that differ from the status quo?

Q27. Which entities would perform the critical functions in tokenised collateral arrangements, and which regulatory frameworks would apply to those functions?

76. Most of the questions in this Call for Evidence aim at helping ESMA in obtaining a better view on how CCP tokenised collateral arrangements could work in practice looking at the specific operational, legal and market risk aspects of tokenised assets. On top of that, ESMA would like to obtain views from respondents on whether the current legal and regulatory framework applicable to collateral posted to CCPs is technologically neutral and “system neutral”. In particular, ESMA is interested in understanding whether existing requirements can accommodate the use of tokenised collateral irrespective of the technology or infrastructure through which it is issued, held, transferred or settled. ESMA would welcome feedback to cover all relevant EMIR requirements (e.g. segregation, investment policy, settlement, etc) as well as other legal texts such as the Settlement Finality Directive.

Q28. Is the current legal and regulatory framework technologically and system neutral, i.e. would it allow for the use of tokenised collateral irrespective of the technology or infrastructure through which it is issued, held, transferred or settled? Are there aspects of it which should be amended or clarified? Please elaborate and identify any provision that while not excluding DLT, presupposes conventional intermediaries, accounts, settlement structures or transfer formalities.

8 Other considerations and tokenisation use-cases

77. While this Call for Evidence focuses on specific legal, operational, market and risk-related aspects of tokenised collateral, ESMA recognises that tokenisation is a rapidly evolving area and that not all relevant considerations may be captured by the questions included in this document. Stakeholders may therefore wish to provide additional information, observations or evidence that could contribute to ESMA's understanding of the potential use of tokenised collateral by CCPs, the broader implications of tokenisation for central clearing, or developments that may become relevant in the future.

Q29. Are there any additional considerations, risks, opportunities, market developments or practical experiences relating to the use of tokenisation in the context of CCP clearing that have not been covered elsewhere in this Call for Evidence? Please elaborate.

78. Beyond collateral mobilisation, there might be other use cases of tokenisation relevant for CCPs. This Call for Evidence focuses primarily on tokenised collateral because it appears to be the most immediate and operationally relevant use case in central clearing. However, tokenisation initiatives in wholesale markets are not limited to collateral and may also affect the broader clearing and post-trade lifecycle.

79. In a CCP environment, possible additional use cases might include the use of tokenisation to streamline post-trade processing (e.g. matching and/or netting), the potential integration of tokenised assets into broader default management processes, or the use of DLT-based infrastructures in areas adjacent to collateral management. It may also be that market participants see little immediate value for CCPs outside tokenised collateral and settlement-related functions. It would therefore be useful to understand whether tokenisation is expected to remain concentrated in collateral-related use cases, or whether stakeholders foresee other developments that could become relevant for CCPs over time.

Q30. Beyond tokenised collateral, are there other use cases of tokenisation that may become relevant for CCPs? If so, which use cases appear the most plausible in the near and medium term?

9 Annexes

9.1 Annex I – List of questions

Q1. Are you currently using, developing or planning to develop any tokenisation solution relevant to collateralisation central clearing? More generally, are you involved in any tokenisation initiatives, projects or arrangements that could help inform ESMA's understanding of the potential use of tokenised collateral by CCPs? Please describe the initiative and its relevance.

When replying to the questions below, please indicate, if necessary, to which tokenisation solution(s) the answer applies.

Q2. Does the taxonomy used by ESMA in this Call for Evidence (“digital twin”, “native”, “hybrid”) correspond to the practice or are there other “models” to consider? How do you see the evolution of these models and what is the role you envisage for tokenised collateral/assets in clearing more generally?

Q3. Is there any specific preferred model for tokenised collateral in the clearing space? If yes, how would this interact with the existing regulatory framework?

Q4. Depending on the preferred model (or models), how would tokenised collateral be handled operationally in a default scenario? Please describe the main steps from posted collateral to liquidity realisation and including any key bottlenecks or dependencies.

Q5. How are (or should be) tokenised collateral arrangements structured from a legal perspective (for example in terms of title transfer or security interest), and do current frameworks (mostly embedded in national laws) provide sufficient clarity as to the nature of the rights involved?

Q6. To what extent may tokenised collateral arrangements ensure that the CCP can exercise its rights over collateral effectively and without delay in the event of default?

Q7. Are there specific legal uncertainties or barriers, including in cross-border contexts, that may affect the enforceability of tokenised collateral?

Q8. How is the applicable law governing property rights determined in relation to tokenised collateral arrangements? Is it determined the same way as for traditional collateral arrangements?

Q9. To what extent do you see legal uncertainties regarding the requirement of possession or control under the FCD? Does control of a wallet or private key have any legal significance for the creation, perfection or enforcement of collateral rights, and if so under which legal framework?

Q10. How are, or should be, segregation and client protection requirements implemented in tokenised collateral arrangements, and does tokenisation affect the level of protection provided compared with conventional models? Please distinguish technical separation from legal separation.

Q11. Do tokenised collateral arrangements support the timely porting of client positions and collateral in a clearing member default? What operational or legal dependencies could affect portability?

Q12. How is the point of finality defined in tokenised collateral arrangements, and how does it relate to both the technical processing of transactions and the legal framework applicable to the system?

Q13. Are there features of tokenised infrastructures or governance arrangements that could adversely affect or on the contrary enhance/support the irreversibility of collateral transfers (and therefore the finality)?

Q14. In arrangements involving multiple layers (such as tokenised systems interacting with traditional infrastructures), how could finality be ensured, including consistency between operational records and legally recognised outcomes?

Q15. For all assets already eligible as CCP collateral, what aspects of their tokenised form would affect the time, process or certainty with which they can be converted into cash, including under stressed conditions? Please explain the factors driving such effects, whether they are expected to be positive or negative, whether they depend on specific operational or technical arrangements, and whether any safeguards or adjustments (including to haircut methodologies) may be warranted.

Q16. Do tokenised collateral arrangements give rise to specific forms of wrong-way risk or concentration risk, in particular where their value or liquidity depends on specific issuers, platforms or counterparties? If so, how should such risks be identified and reflected in collateral eligibility, liquidity assessment or haircut calibration?

Q17. What factors are most relevant for assessing the liquidity of tokenised collateral, including for both securities and tokenised cash?

Q18. Do tokenised arrangements (e.g. faster settlement or automation, automation, programmability or continuous processing) have broader implications for liquidity risk, including intraday needs or procyclicality?

Q19. For assets already eligible as CCP collateral, would tokenisation affect the ability of CCPs to manage and invest collateral in accordance with EMIR requirements? Please explain.

Q20. Does this description accurately capture the main operational and cyber risks associated with tokenised collateral in a CCP context? If not, which additional risks should be considered? Please elaborate detailing the taxonomy of tokenised-specific risks that should be incorporated into CCP risk management approaches.

Q21. Which components of tokenised collateral arrangements should be regarded as critical dependencies for supervisory purposes, and why?

Q22. What forms of testing, assurance (including but not limited to potential fallback arrangements) and scenario analysis (in particular potential new extreme but plausible

scenarios) would be necessary before tokenised collateral arrangements could be relied upon more broadly by CCPs?

Q23. Does this description accurately capture the main interoperability challenges associated with tokenised collateral in a CCP context? In particular, would a lack of interoperability risk fragmenting tokenised collateral across platforms, infrastructures or CCPs?

Q24. Are common market practices, technical standards or regulatory approaches likely to be needed to support interoperability and avoid the emergence of closed or fragmented tokenisation models?

Q25. Could tokenisation lead to new market structure dependencies or concentration risks? If so, in what form?

Q26. How do you expect market structure to be impacted by broader usage of tokenised collateral? What sort of risks could be envisaged? How would that differ from the status quo?

Q27. Which entities would perform the critical functions in tokenised collateral arrangements, and which regulatory frameworks would apply to those functions?

Q28. Is the current legal and regulatory framework technologically and system neutral, i.e. would it allow for the use of tokenised collateral irrespective of the technology or infrastructure through which it is issued, held, transferred or settled? Are there aspects of it which should be amended or clarified? Please elaborate and identify any provision that while not excluding DLT, presupposes conventional intermediaries, accounts, settlement structures or transfer formalities.

Q29. Are there any additional considerations, risks, opportunities, market developments or practical experiences relating to the use of tokenisation in the context of CCP clearing that have not been covered elsewhere in this Call for Evidence? Please elaborate.

Q30. Beyond tokenised collateral, are there other use cases of tokenisation that may become relevant for CCPs? If so, which use cases appear the most plausible in the near and medium term?

9.2 Annex II – Definitions

80. **Asset-referenced token (ART)** as defined in Article 3(1) of Regulation (EU) 2023/1114 (MiCA), an asset-referenced token is a type of crypto-asset that is not an e-money token and that purports to maintain a stable value by referencing another value or right, or a combination thereof, including one or more official currencies.
81. **Central Counterparty (CCP)** as defined in Article 2(1) of Regulation (EU) No 648/2012 (EMIR), CCP is a legal person that interposes itself between the counterparties to contracts traded on one or more financial markets, becoming the buyer to every seller and the seller to every buyer.
82. **Central Securities Depository (CSD)** as defined in Article 2(1) of Regulation (EU) No 909/2014 (CSDR), a CSD is a legal person that operates a securities settlement system and provides at least one other core service specified in CSDR.
83. **Clearing** as defined in Article 2(3) of EMIR means the process of establishing positions, including the calculation of net obligations, and ensuring that financial instruments, cash, or both, are available to secure the exposures arising from those positions.
84. **Consensus mechanism** as defined in Article 3(1)(3) of Regulation (EU) 2023/1114 (MiCA) and Article 2 of Regulation (EU) 2022/858 (DLT Pilot Regime), a consensus mechanism is the rules and procedures by which an agreement is reached among DLT network nodes that a transaction is validated.
85. **Crypto-asset** is a digital representation of a value or a right that is able to be transferred and stored electronically using distributed ledger technology or similar technology.
86. **Digital twin** for the purposes of this Call for Evidence, refers to a tokenised representation of a traditionally issued and held asset, where the underlying asset continues to exist and be recorded within conventional custody, settlement or CSD infrastructures. The token acts as a representation of, or operational layer associated with, the underlying asset rather than constituting the primary legal existence of the asset itself.
87. **Distributed Ledger Technology (DLT)** as defined in Regulation (EU) 2022/858 (DLT Pilot Regime), means a technology that enables the operation and use of distributed ledgers. A distributed ledger is an information repository that keeps records of transactions and that is shared across, and synchronised between, a set of DLT network nodes using a consensus mechanism.
88. **DLT financial instrument** as defined in Regulation (EU) 2022/858 (DLT Pilot Regime), is a financial instrument within the meaning of MiFID II that is issued, recorded, transferred and stored using distributed ledger technology.

89. **DLT market infrastructure** as defined in Regulation (EU) 2022/858, refers collectively to DLT multilateral trading facilities (DLT MTFs), DLT settlement systems (DLT SSs) and DLT trading and settlement systems (DLT TSSs) operating under the DLT Pilot Regime.
90. **E-money token (EMT)** as defined in Article 3(1)(7) of Regulation (EU) 2023/1114 (MiCA), an e-money token is a type of crypto-asset that purports to maintain a stable value by referencing the value of one official currency.
91. **EMIR** Regulation (EU) No 648/2012 of the European Parliament and of the Council on OTC derivatives, central counterparties and trade repositories, commonly referred to as the European Market Infrastructure Regulation.
92. **Native tokenised asset** for the purposes of this Call for Evidence, refers to an asset that is issued, recorded, held and transferred directly on a distributed ledger, such that the token constitutes the primary representation of the asset.
93. **Settlement** as defined in Article 2(7) of CSDR, settlement means the completion of a securities transaction where it is concluded with the aim of discharging the obligations of the parties through the transfer of cash or securities, or both.
94. **Settlement finality** for the purposes of this Call for Evidence, settlement finality refers to the point at which a transfer, settlement obligation or collateral movement becomes legally effective, irrevocable and protected against subsequent challenge, including in insolvency proceedings. Within the EU framework, this concept is closely linked to the protections under Directive 98/26/EC on settlement finality in payment and securities settlement systems.
95. **Token** for the purposes of this Call for Evidence, a token means a digital representation of an asset, right, claim or value recorded on a distributed ledger and capable of being transferred or controlled through that ledger.
96. **Tokenisation** for the purposes of this Call for Evidence, tokenisation means the digital representation of an asset or right on distributed ledger technology (DLT) or a similar technology, such that transfer and lifecycle events can be recorded and, in some cases, executed in a digital and potentially automated manner.
97. **Tokenised cash** for the purposes of this Call for Evidence, tokenised cash refers to a tokenised instrument intended to perform the economic function of cash or a settlement asset. Depending on its legal structure, this may include tokenised deposits, e-money tokens, asset-referenced tokens referencing a currency, or other tokenised representations of monetary claims.
98. **Tokenised collateral** for the purposes of this Call for Evidence, tokenised collateral means collateral represented, recorded or mobilised through a tokenised arrangement, whether through a digital twin model or a native tokenised asset model.

99. **Tokenised security** for the purposes of this Call for Evidence, a tokenised security means a financial instrument represented or issued using distributed ledger technology, including both digital twin representations of traditionally issued securities and DLT financial instruments issued natively on a distributed ledger.